

Course Syllabus

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EE 122: Introduction to Control Systems

Instructor: Ayush Pandey (meet with me 1-on-1 using this [URL](https://outlook.office.com/bookwithme/user/01b63bdd7fca46e09e7832292840538e@ucmerced.edu/meetingtype/iseQLcF3dkS9_VQh-6DdLw2?anonymous&ep=mink)  [_](https://outlook.office.com/bookwithme/user/01b63bdd7fca46e09e7832292840538e@ucmerced.edu/meetingtype/iseQLcF3dkS9_VQh-6DdLw2?anonymous&ep=mink)(https://outlook.office.com/bookwithme/user/01b63bdd7fca46e09e7832292840538e@ucmerced.edu/meetingtype/iseQLcF3dkS9_VQh-6DdLw2?anonymous&ep=mink)); course Box)

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 **Class Discussions:** Mon and Wed from 6:00 PM - 7:50 PM at COB2 170

 **Ayush's office hours:** Mondays from 5pm to 6pm in SE2 381

 **Lab 02L:** Tue (pre-lab discussion and office hour) and Thu (experimental lab) from 1:30 PM - 3:20 PM

 **Lab 04L:** Mon (pre-lab discussion and office hour) and Wed (experimental lab) from 11:30 AM - 1:20 PM

 GitHub: <https://github.com/ee-ucmerced/ee122-control-systems>  [_](https://github.com/ee-ucmerced/ee122-control-systems)(https://github.com/ee-ucmerced/ee122-control-systems)

Textbooks and Resources:

- (primary text) Åström, Karl Johan, and Richard Murray. "Feedback Systems: An Introduction for Scientists and Engineers". Princeton University Press, 2021.
 - Official copy is freely available online: https://fbswiki.org/wiki/index.php/Feedback_Systems:_An_Introduction_for_Scientists_and_Engineers  [_](https://fbswiki.org/wiki/index.php/Feedback_Systems:_An_Introduction_for_Scientists_and_Engineers)(https://fbswiki.org/wiki/index.php/Feedback_Systems:_An_Introduction_for_Scientists_and_Engineers)
 - Associated Python package (equivalent of MATLAB Control Systems Toolbox) that the book has examples for is called python-control: <https://python-control.readthedocs.io/en/0.10.2/>  [_](https://python-control.readthedocs.io/en/0.10.2/)(https://python-control.readthedocs.io/en/0.10.2/)
- (supplementary) B. P. Lathi and Roger Green. "Signal Processing and Linear Systems"

Second Edition. ISBN: 9780190299040. E-book ISBN: 9780190857691

Why take EE 122?

EE 122 / ME 141 is an introductory course and an entry point into the vast area of control systems and control theory. This course is perhaps the most interdisciplinary course you'll come across. The concepts in this course are directly applicable to many areas: electrical engineering (ex: how do you maintain a constant DC voltage, when needed, under disturbances?), mechanical engineering (ex: how does cruise control work?), aerospace engineering (ex: how does one navigate a drone to a desired location?), bioengineering (ex: how can we genetically engineer a crop to increase production?), biological sciences (ex: how does the brain use eyes to control hand movement?), economics (ex: how to make the most money!?). These questions are not hypothetical or far-reaching. In fact, these are questions that you can immediately answer by direct application of concepts in this course!

About EE 122 (catalog description)

Introduces basic concepts of single-input single-output closed-loop control systems in both time and frequency domains. Controller designs and concepts include root-locus method, frequency-domain synthesis, PID tuning, lead-lag compensators, internal model principle, and design tradeoffs. These methods are used to reach desired system behavior including stability, stability margins, sensitivity, and time/frequency-domain specifications. Practical issues like nonlinearities in control systems and describing function analysis; sampling data control systems and digital control designs will be introduced. State-space models, controllability and observability and other modern multi-input multi-output control systems concepts will be introduced briefly.

Course Learning Outcomes

- CLO 1: Apply advanced skills to analyze dynamics of linear systems (PLO 1)
- CLO 2: Apply the knowledge of matrix theory to analyze linear systems and to design linear controls for regulating the dynamics of the system (PLO 1)
- CLO 3: Gain a thorough understanding of the theory of feedback controls and stability (PLOs 1, 7)
- CLO 4: Apply methods of control design including root locus, frequency domain, state space designs (PLO 1)
- CLO 5: Understand the concepts of controllability and observability (PLO 1)
- CLO 6: Apply the concepts of optimization to design optimal controls (PLO 1)
- CLO 7: Apply control theories, control methods, and critical thinking skills to control problems of

engineering systems (PLO 1, 7)

Course prerequisites:

MATH 024 and (ME 140 or EE 102)

Course policy and expectations:

1. **Lectures** will be in-person and will use the whiteboard in the classroom. In-class activities will include problem solving, group discussions, computer simulations, and pop quizzes. Lecture notes will be provided after the class so you can catch up on any missed material and compare your own notes to what was covered in the class.
2. **Participation:** Participation points will be made up of two types of assignments: in-class activities during the lectures + reading assignments that are effectively for discussion attendance. In-class activities that are online can be completed until midnight of the day of the class. In-class activities that are in-person can only be completed during the lecture time and cannot be made up later. Missed reading assignments cannot be made up later. Around 10-15% of participation assignments will be dropped from the grade computation so you won't lose any points towards the final grade if you only miss a handful of lectures and discussion sections.
3. **Labs:** Labs will be conducted in SE1 172 and will involve hands-on experiments with the equipment in the lab. A lab report will be due at the end of each lab section. Missed labs can be made up only 1-week after it was originally due. The lab reports must reflect your original work and only experimentally observed data must be reported. Plagiarism/fabrication in the lab reports will be strongly penalized.
4. **Extensions:** For problem sets, you have **three free (no questions asked) extension requests**. You can make up lab assignments for free up until 1 week but you must find a time when you can access the equipment. If the lab is busy and the equipment is unavailable, you may not be able to make up missed labs. Late work on problem sets is still eligible for partial credit: 7% late penalty for every day the assignment is late up to a maximum of 49% late penalty. The latest you can submit an assignment is 7 days after the regular deadline.
5. **Academic honesty:** All work you turn in must be your own (including the code). Collaborations are encouraged but the final work that you submit must be your own. Asking for help in understanding a concept, learning from discussions online, and asking ChatGPT (see more below) for explanations are all OK — but, you must complete your own work after you have received this help and understood the concepts. If plagiarism is detected, you will be given a warning. On repeat offense, an action will be taken according to the university academic honesty policy [PDF \(https://osrr.ucmerced.edu/sites/osrr.ucmerced.edu/files/documents/academic_honesty_-_800.pdf\)](https://osrr.ucmerced.edu/sites/osrr.ucmerced.edu/files/documents/academic_honesty_-_800.pdf).

6. **AI use:** You are allowed to use AI tools to help with your learning and for drafting boilerplate code for your assignments. However, you must fully understand all work you turn in (including code) and you should be able to make changes to it as needed. You must acknowledge in your submission that an AI tool was used and indicate how AI was used. Acknowledging AI use will never lead to penalties.
7. **Class Conduct and Community:** Remember that contributions from each of us can help in building a respectful, courteous, and an intellectually stimulating class environment. Language or behavior that prevents any student to participate fully in class is not acceptable. It is important to remain open to each other's thinking and engage in rigorous, challenging discussion about issues of shared concern. Please refer to the full student conduct policy [PDF \(https://osrr.ucmerced.edu/sites/osrr.ucmerced.edu/files/documents/code_of_conduct_600_and_700.pdf\)](https://osrr.ucmerced.edu/sites/osrr.ucmerced.edu/files/documents/code_of_conduct_600_and_700.pdf).
8. **Accommodations:** All accommodation requests are welcome. You can submit an accommodation letter through the Student Accessibility Services for this course. However, if you are unable to do so, you can directly discuss any accommodations that you may need with the instructor and we will do our best to provide a supportive learning environment for you.
9. **Support and Services for Undocumented Students:** In this time of uncertainty, the university has reaffirmed their commitment to supporting undocumented students. I would like to highlight these resources in the course syllabus to promote inclusion and belonging as we embark on this learning experience together, despite the uncertain and oppressive situation around us. At UC Merced, the Monarch Center offers support to undocumented and mixed-status students through resources, advocacy, and co-curricular programs aimed at promoting academic success and personal growth. Go here to learn more: <https://undoc.ucmerced.edu/>
10. UC Merced is committed to providing an **equal opportunity environment** for all students and employees that remains free of all forms of discrimination, harassment, and exploitation. Discrimination and harassment based on race, color, religion, creed, sex, national origin, age, disability, veteran status, or sexual orientation is a violation of state and federal law and/or University policy and will not be tolerated. Retaliation against any person who complains about discrimination is also prohibited. If you witness or experience any form of harassment, please seek support and guidance. For more information, please visit <https://ophd.ucmerced.edu/> (<https://ophd.ucmerced.edu/>)
11. **Emailing/direct messaging on CatCourses:** If the syllabus does not answer your question, please reach out to the TA in the lab section. If your question needs urgent attention, you can send an email to the TA or the instructor (make your best judgement on who might answer your question best and quickest). Please **mention the course number in the subject** of any email you send.

Assessment structure:

1. All in-class activities and reading assignments during the discussion section are graded for

participation: 10% of your total grade (around 4-5 lowest scoring assignments will be dropped).

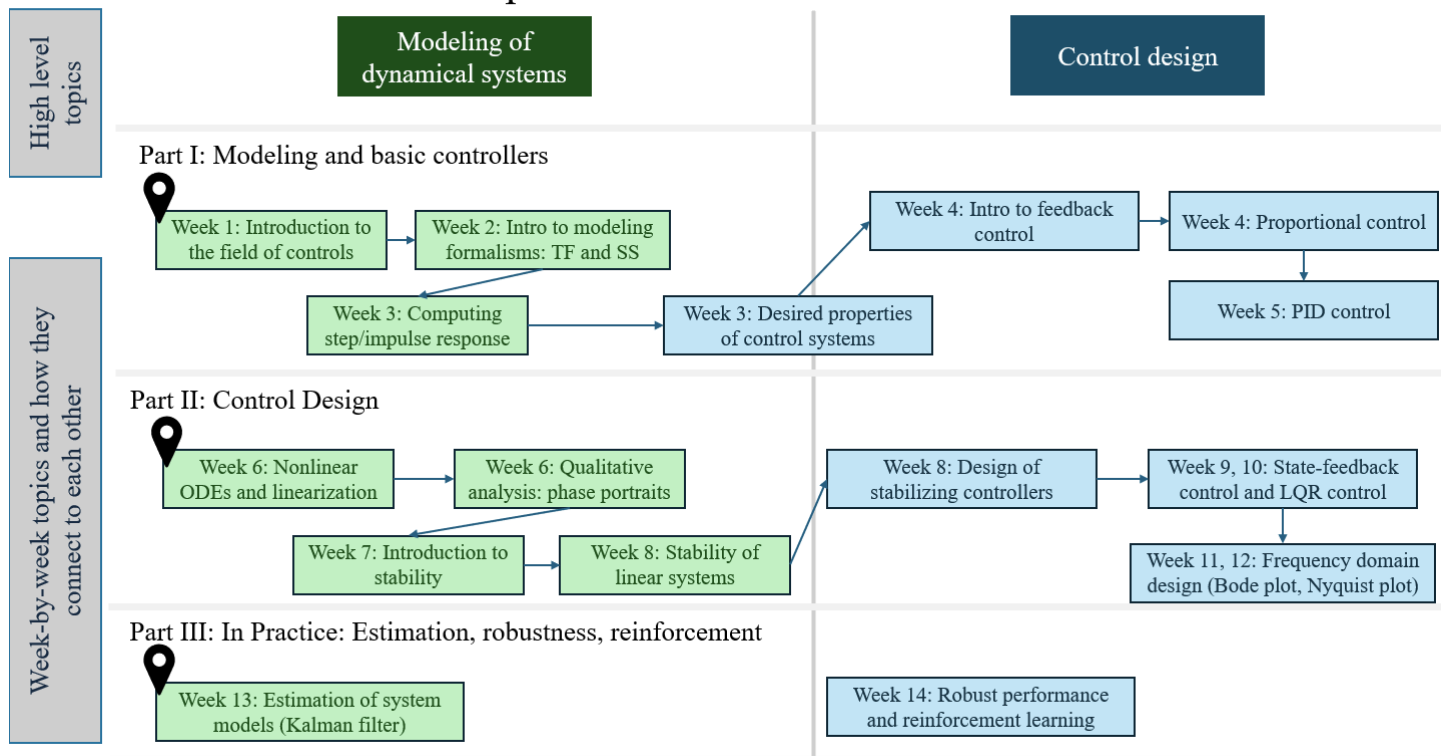
2. All lab reports are graded for correct completion: 20% of your total grade (two lowest scoring labs will be dropped)
3. All problem sets are graded for correctness: 20% of your total grade (two lowest scoring problem sets will be dropped)
4. Summative assessment consists of two exams and one project: midterm (15% of your total grade), final (20% of your grade), and project (15% of your total grade)

Grading Scheme

Letter Grade	Range
A+	100% to 99%
A	< 99% to 95%
A-	< 95% to 90%
B+	< 90% to 87%
B	< 87% to 83%
B-	< 83% to 80%
C+	< 80% to 77%
C	< 77% to 73%
C-	< 73% to 70%
D+	< 70% to 67%
D	< 67% to 63%
D-	< 63% to 60%
F	< 60% to 0%

Course timeline and plan

EE 122 / ME 141 Course Map



Weekly Course Schedule

Date	Topics	Related assignments and due dates	Reading / Other notes
01/21	Introduction to feedback systems	None	FBS Chapter 1
01/26	Modeling dynamical systems	Reading 1 during M-T discussion	
01/28	Canonical control example: Inverted pendulum	Problem set 1 due on Wed Lab 1 during W-Tr lab	FBS Chapter 3

02/02	Transfer functions and system properties	Reading 2 during M-T discussion	
02/04	State space representations of dynamical systems	Problem set 2 due on Wed	FBS Chapter 3
	Properties of step response and impulse response	Lab 2 during W-Tr lab	
02/09	Properties of systems: poles and stability	Reading 3 during M-T discussion	FBS Chapter 4 and some topics on block diagrams and transfer functions from Chapter 9
02/11	Simple controllers: Proportional, and how they impact system properties	Problem set 3 due on Wed	
		Lab 3 during W-Tr lab	
02/18	PID controllers	Reading 4 during M-T discussion	
		Problem set 4 due on Wed	Topics from Chapter 11
		Lab 4 during W-Tr lab	
02/23		Reading 5 during M-T discussion	
02/25	Dynamical properties of systems with controllers (in state-space)	Problem set 5 due on Wed	Chapter 5
		Lab 5 during W-Tr lab	
03/02	Stability of dynamical systems	Reading 6 during M-T discussion	
03/04	03/04: Midterm	Problem set 6 (midterm practice exam) due on Wed	Chapter 5 and 6
		No labs this week.	
		Midterm of 03/04	

03/09	Stability of linear systems		
03/11	Diagonalization and introduction to state-feedback control		
03/16	State-feedback control		
03/18	LQR control		
04/01	LQR control		
04/03	Time-domain control methods		
04/08	Pole placement		
04/10	Bode plots		
04/15	Bode plots for control design		
04/17	Nyquist plots		
04/22	Frequency domain methods for control		
04/24			
04/29	Estimation of systems: Kalman filtering		
05/01	Robust control		
05/06	Reinforcement learning for control		Final project due
05/08	Course Review		
05/12	Final exam		

Course Summary:

Date	Details	Due
		